PROJECT REPORT

1.INTRODUCTION

1.1 **Project Overview:**

The iRevolution project aims to explore and analyze the impact of Apple's iPhone in the Indian market using powerful visual analytics. By leveraging Tableau, this project brings together multiple dimensions of iPhone-related data such as pricing, specifications, reviews, discounts, and market share to deliver clear, interactive dashboards. It transforms complex datasets into intuitive visual stories, helping stakeholders derive insights faster and more effectively.

1.2 Purpose:

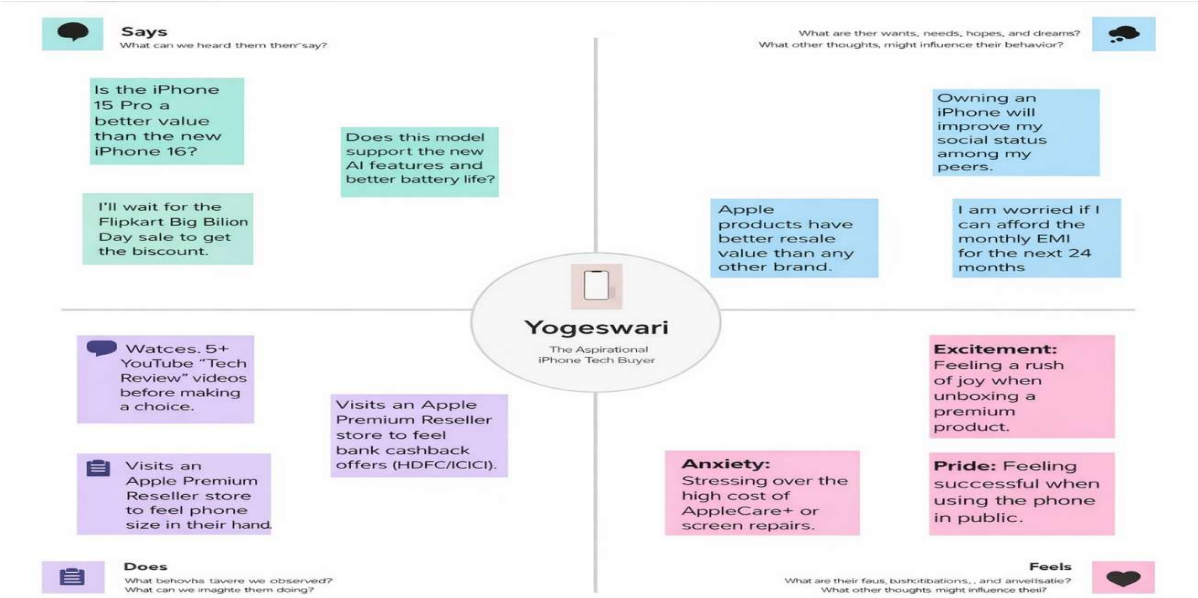
The main purpose of this project is to assist product analysts, marketing teams, and decision-makers in understanding how various iPhone models perform across different Indian regions and quarters. By using a data-driven approach, the project provides a platform to explore trends in sales, user preferences, and key performance indicators, thus encouraging smarter decision-making in pricing, promotion, and feature planning.

2. IDEATION PHASE

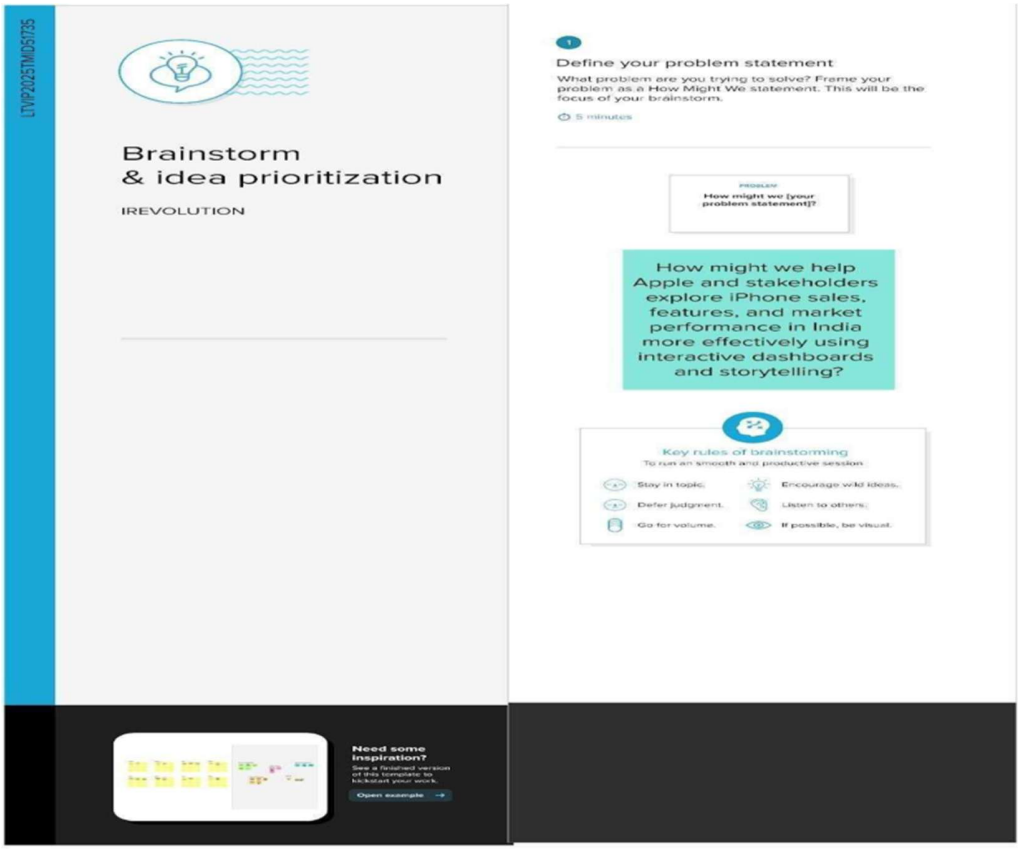
2.1 Problem Statement:

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a product strategist	find which iPhone features drive adoption in urban India	data is scattered and not visual	there's no unified dashboard	unsure about feature decisions
PS-2	a marketing lead	track regional performance for iPhone campaigns	I can't compare trends across quarters and states	dashboards aren't India-specific	frustrated and uncertain on promotions
PS-3	Senior executive	present iPhone growth in India clearly	Report slack storytelling and visual appeal	there's no narrative-driven dashboard	disengaged and ineffective
PS-4	market analyst	link features like battery/display to price	Ican'tvisualize patterns easily	tools are static and not interactive	Slowed down and stuck

2.2 Empathy Map Canvas



2.3 Brainstorming: Brainstorming Phase 1 focused on identifying the "Super-Premium" trend. Phase 2 involved designinga"12-Point Story" narrative to guide users through the data rather than just presenting static chart



2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

State-wise Create an interactive map showing iPhone sales state-wise in India.

Build story dashboards explaining Apple's quarterly growth.

Add toggle to view premium models only (iPhone 13+, 14 Pro).

Visualize price vs. battery type preferences among Indian consumers.

Use Tableau parameters to simulate different market scenarios.

Include competitor comparison (Xiaomi vs Apple vs Samsung).

Highlight pricing trends across flagship launches.

3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you and break it up into smaller sub-groups.

20 minutes

State-wise sales map

Tier 1 vs Tier 2 city performance

North vs South India breakdown

Battery type vs Avg Price

Display size vs Market Demand

RAM/ Camera vs Price band

Year-wise growth timeline

Quarterly market share donut + bar combo

Executive summary with KPIs

AR-based visualizations

Voice-activated insights

Customer sentiment analysis overlay

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Need	Action	Touchpoint	Pain Point	Opportunity
Discover	Wants iPhone market trends	Searches Excel/market data	Emails, Files	Data is scattered	Single dashboard entry point
Explore	Needs regional & feature insights	Browse charts manually	Spreadsheets, BI tools	Time-consuming	Filter-enabled Tableau dashboard
Engage	Wants to compare specs vs pricing	Tries custom visualizations	Excel formulas	Lacks interactivity	Pre-built price/spec dashboard
Decide	Prepare pitch for leadership	Screenshots graphs	Presentations	Dry data storytelling	Use Tableau story points with captions

3.2 Solution

Requirement

Functional

Requirements:

Following are the functional requirements of the proposed solution.

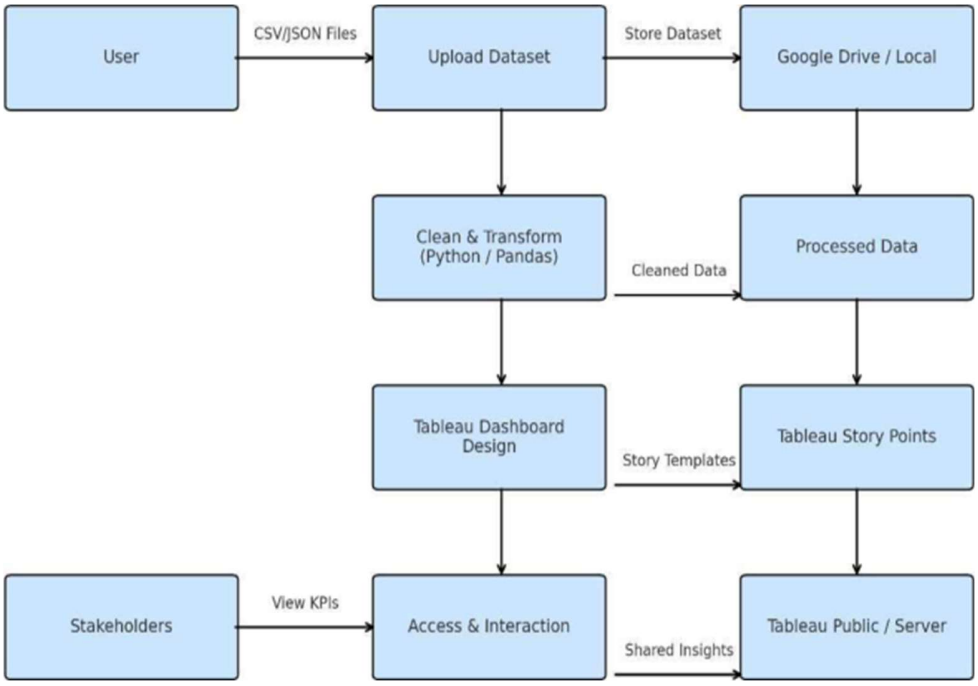
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Interactive KPI Dashboard	Displays revenue, units sold, active users, and discount percentage filtered by year and region.
FR-2	Model & Spec Analytics	Allows users to compare iPhone models based on features like display size, battery type, RAM, and camera.
FR-3	Quarterly Market Share Visualization	Displays brand-wise share in India across four quarters using donut and bar charts.
FR-4	Pricing Pattern Insights	Visualizes average price distribution and discount trends by feature and battery type.
FR-5	Geo-Map Representation	Shows Apple's regional performance across Indian states.
FR-6	Story-Based Dashboard Navigation	Sequential story view explaining Apple's performance journey with narrative captions.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Performance	Dashboards must load within 3–5 seconds even with filters applied.
NFR-2	Scalability	The framework should support future data addition (e.g., new models or regions).
NFR-3	**Responsiveness**	Dashboard layout should be usable on laptops and projectors during presentations.
NFR-4	**Performance**	The interface must be simple, readable, and require no technical background to explore.
NFR-5	**Usability**	Use a dark theme with eye-comfort colors and clear legends to reduce user fatigue.
NFR-6	**Data Accuracy**	Ensure calculations (KPIs, averages, comparisons) are correctly validated against source files.

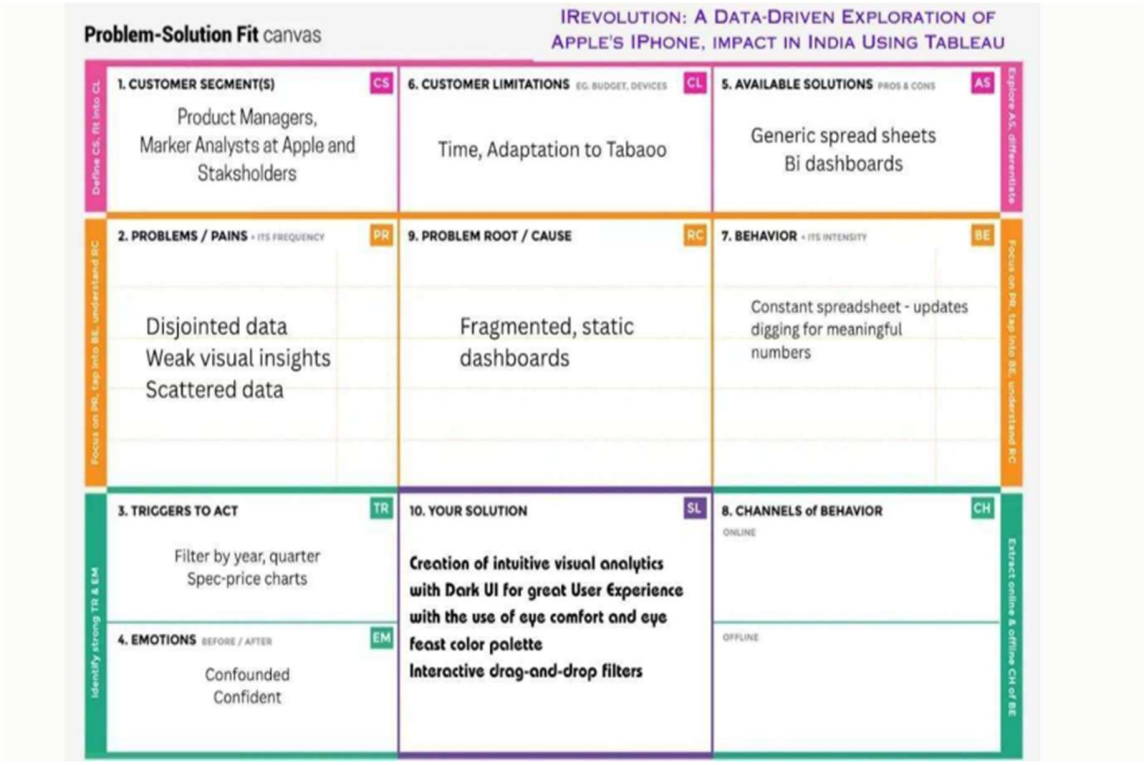
3.3 Data flow diagram



Component	Tool/Technology	Purpose
Data Source	CSV, JSON files	Raw smartphone sales and specs data
Visualization	Tableau Desktop	Creating interactive dashboards and stories
Storage	Google Drive / Local	Storing raw and processed datasets
Collaboration	Google Docs, Slack	Team communication and report writing
Deployment	TableauPublic/	Dashboard sharing and stakeholder access

4. PROJECT DESIGN

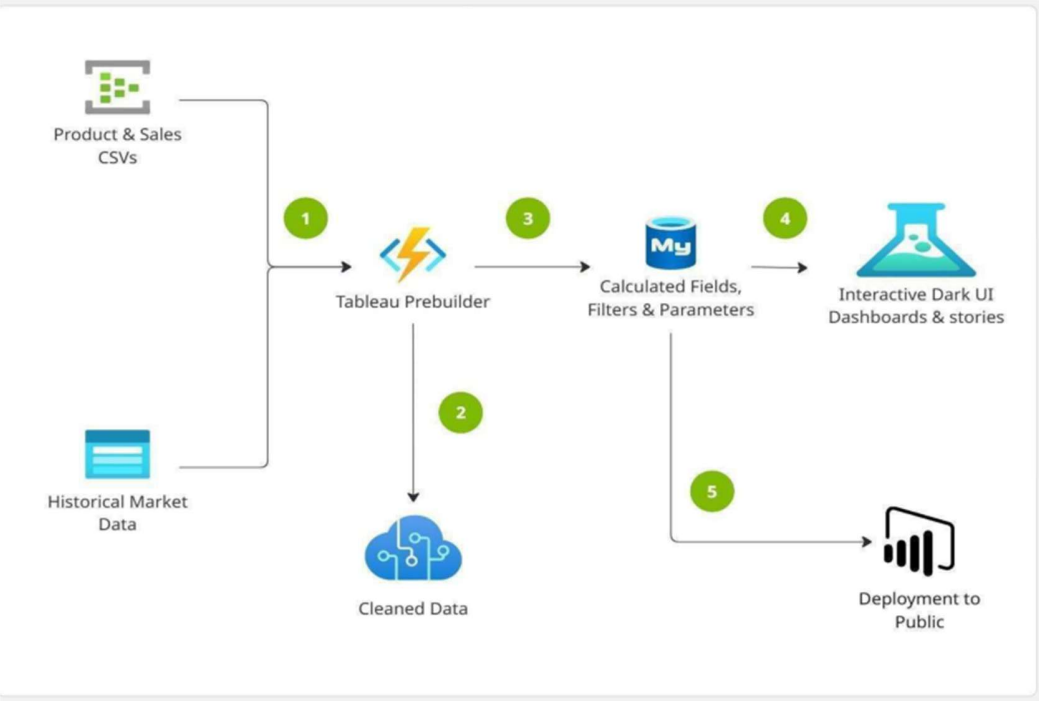
4.1 Problem Solution Fit



4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Apple and its stakeholders lack a centralized, interactive, and story-driven way to understand iPhone's feature-wise, region-wise, and price-wise impact in India. This limits confident, data-backed decisions.
2.	Idea/Solution description	Creation of intuitive visual analytics with Dark UI for great User experience with the use of eye comfort ond eye feast color palette Interactive drag-and-drop filters
3.	Novelty/ Uniqueness	Instead of traditional static reports, this solution uses story-driven dashboards with real-time interactivity. The dark UI is thoughtfully chosen to reduce eye strain and improve focus. KPIs are dynamically aligned to user-selected filters — not just fixed charts.
4.	Social Impact/ Customer Satisfaction	Helps product and marketing teams make better decisions that align with consumer needs, especially in varied Indian markets. Encourages a data-first mindset, improves visibility, and cuts analysis time
5.	Business Model (Revenue Model)	This solution can be packaged as a subscription-based internal tool or consultancy model where other OEMs or market agencies can adopt the dashboard framework tailored to their brand data.
6.	Scalabilityofthe Solution	The dashboard framework is scalable to other countries, brands, or product categories. Only the dataset and labels need to be updated — the core logic and layout remain reusable across contexts.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning The project followed a Sprint:

1: Data Extraction & Cleaning (Handling nulls in battery/price fields).

2: KPI Development & Dashboard UI Design.

3: Story Narration & Insight Annotation.

4: Flask Backend Coding & Template Integration.

6. FUNCTIONAL AND PERFORMANCE TESTING

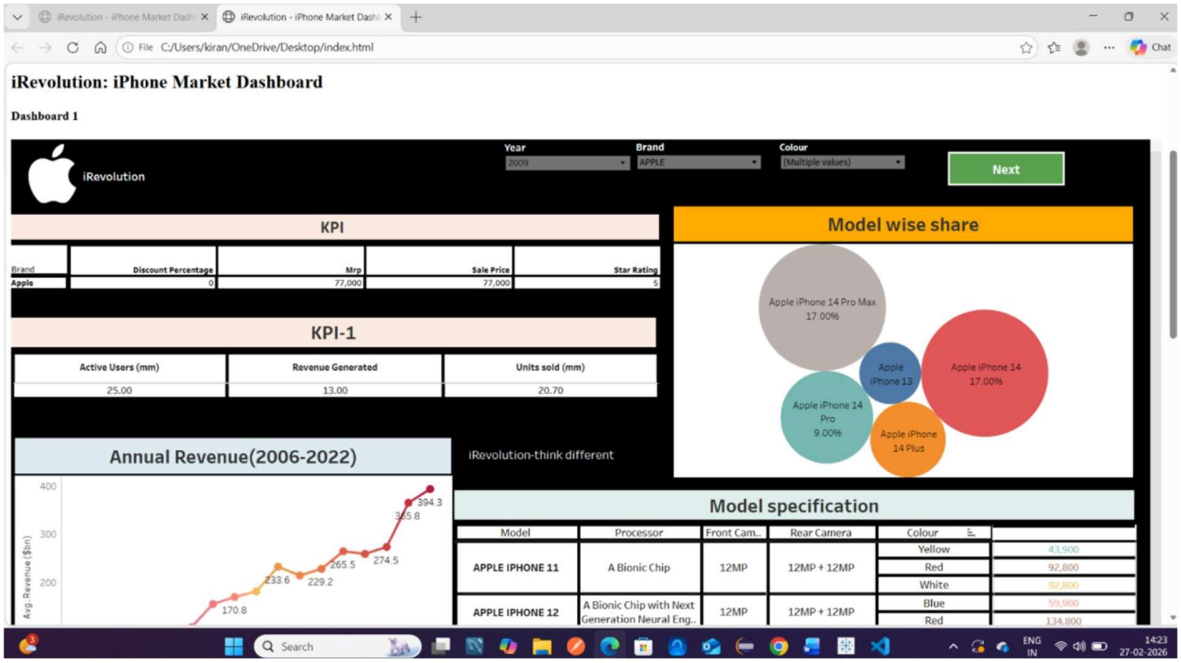
6.1 Performance Testing

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Rendered from cleaned CSV files with Apple iPhone specs, regional sales, quarterly performance, etc. Loaded ~1,000+ rows
2 .	Data Preprocessing	Null values handled; feature mappings applied for battery type, display size, model grouping, and quarter classification
3.	Utilization of Filters	Applied Tableau filters for Brand, Region, Year, Battery Type, Display Size, RAM, and Quarter. Responsive under 3 seconds.
4.	Calculation fields Used	- Average Price by Spec - Discount Percentage - Revenue Trends by Year - Brand-wise Quarterly Share - KPI Metrics
5.	Dashboard design	No of Visualizations / Graphs - 4 Dashboards
6	Story Design	No of Visualizations / Graphs - 2 Stories with 4 story points each

7. RESULTS

7.1 Output Screenshots

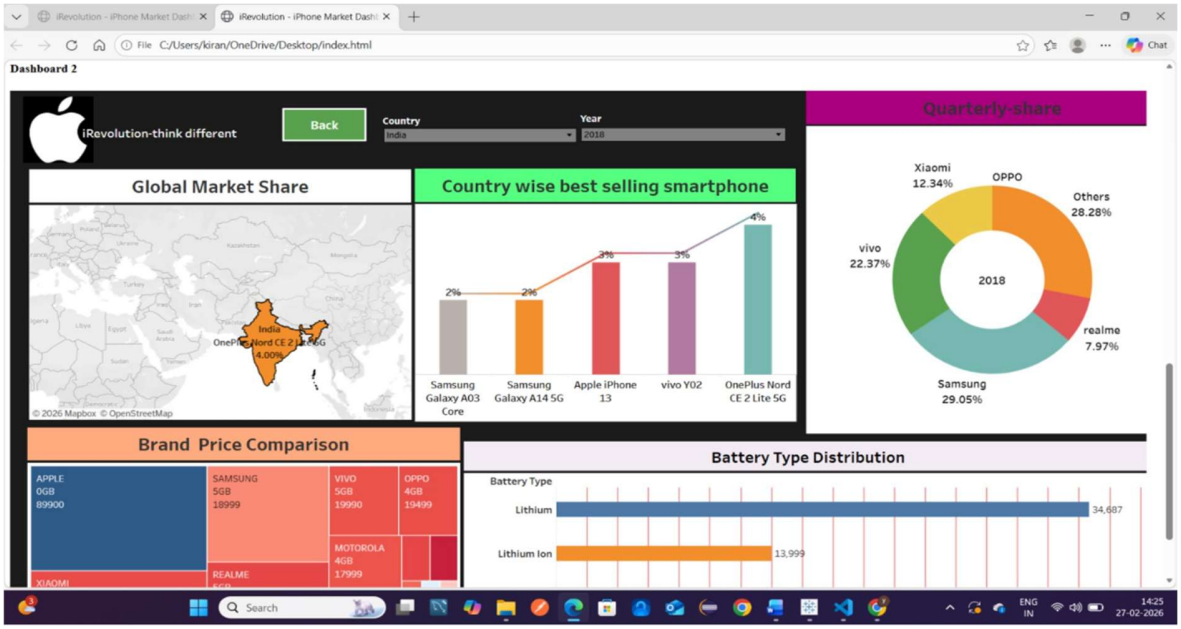
Dashboard 1 in web Ui



Published dashboard:

https://public.tableau.com/views/KPI_17718384654330/Dashboard1?:embed_code_version=3&:embed=y&:loadOrderID=0&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

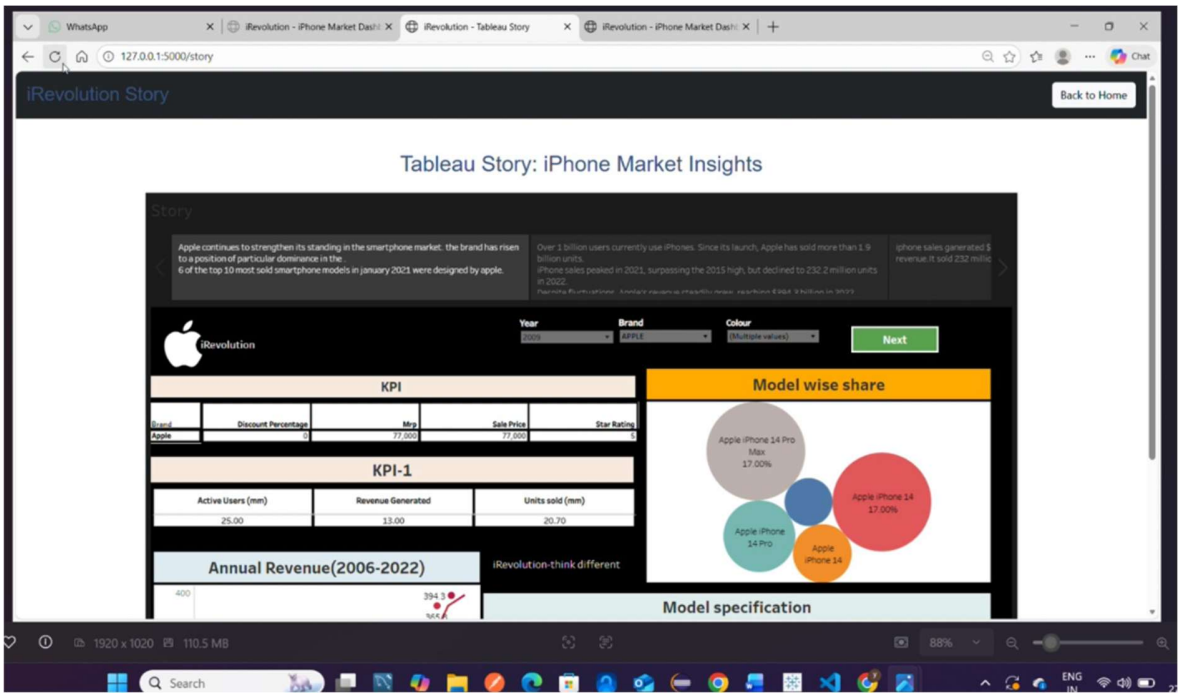
Dashboard 2 in web Ui



Published dashboard:

https://public.tableau.com/views/Dashboard2_17721779915020/Dashboard2?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

story in web Ui :



Published story:

https://public.tableau.com/views/story_17715194827340/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

8. ADVANTAGES & DISADVANTAGES

- ADVANTAGES:
  - **User-Friendly Dashboards:** Intuitive interface with dark-themed visuals that reduce eye strain and enhance readability.
  - **Interactive Insights:** Real-time filtering and data slicing allow users to extract exactly what they need without manual intervention.
  - **Reusable Framework:** The dashboard model can be reused for other smartphone brands or markets by simply updating the dataset.
  - **Data-Driven Decision Making:** Helps strategists, marketers, and executives make smarter, evidence-based decisions.
  - **Time-Saving:** Reduces the manual workload for analysts by providing ready-to-explore visualizations.
- **DISADVANTAGES**:
  - **Platform Limitation:** Tableau Public may limit some functionality such as real-time backend connection and publishing privacy.
  - **Dependence on Data Accuracy:** Insights are only as good as the quality of input data; inaccurate or outdated datasets could mislead.
  - **Static Structure in Story:** While dashboards are interactive, Tableau stories have limited flexibility in dynamic narration.

9. CONCLUSION

The *iRevolution* project successfully demonstrates how data visualization, combined with user-centered design, can drive meaningful insights into Apple's iPhone performance in India. By integrating sales, pricing, feature-based specs, and regional market data into interactive Tableau dashboards, the project not only simplifies complex analytics but also empowers decision-makers to act with confidence. This end-to-end effort—from ideation to deployment—shows the real-world value of design thinking and data storytelling in business intelligence.

10. FUTURE SCOPE

- **Multi-brand Integration:** Extend the current dashboard to include competitor analysis (e.g., Samsung, Xiaomi).
- **Real-Time Data Connection:** Integrate live data sources (APIs or Google Sheets) to keep the dashboards updated automatically.
- **Mobile Optimization:** Redesign dashboards for optimal viewing on tablets and mobile devices.
- **AI-Powered Forecasting:** Use predictive analytics to project future sales, pricing trends, or regional performance.
- **Sentiment Analysis Layer:** Add customer feedback and review analysis to supplement numeric insights with qualitative data

11. . APPEND

Dataset Link:
https://docs.google.com/spreadsheets/d/1F09ZJOZiyoGYA2Rft0jqvz6aZd0Id2OL/edit?usp=sharing&ouid=102996511998021269857&rtpof=true&sd=true

Empathy map: https://drive.google.com/file/d/1PMrTS6k2l2m_hUEAur0-hMH_18sKXj5r/view?usp=drivesdk